Major Project Report

on

"MediVoice"

Submitted in Partial fulfillment for the award of degree of Bachelor of Technology in Computer Science and Engineering

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Session: 2025-26

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Session: 2025-26

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

CERTIFICATE

This is to certify that the work embodied in this Project Report entitled as **"MediVoice – AI Doctor Website"** being Submitted by **Palak Bhargav (0111CS221144), Naina Gupta (0111CS221135), Chetan Pawar (0111CS221073),** in partial fulfillment of the requirement for the award of **"Bachelor of Technology"** in Computer Science & Engineering discipline to Rajiv Gandhi Proudyogiki Vishwavidyalaya, Bhopal (M.P.) during the academic year 2025-26 is a record of bonafide piece of work, carried out under my supervision and guidance in the Department of Computer Science & Engineering, Technocrats Institute Of Technology, Bhopal.

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Approved by AICTE New Delhi & Govt. of M.P.

Session: 2025-26

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

CERTIFICATE OF APPROVAL

This Project **"MediVoice – AI Doctor Website"** being submitted by **Palak Bhargav (0111CS221144), Naina Gupta (0111CS221135), Chetan Pawar (0111CS221073)**, has been examined by me & hereby approve for the partial fulfilment of the requirement for the award of **"Bachelor of Technology"**in **"Computer Science & Engineering"** for which it has been submitted. It is understood that by this approval the undersigned do not necessarily endorse or approve any statement made, opinion expresses or conclusion draw there in, but the Project only for the purpose for which it has been submitted.

Internal Examiner

Date:

CANDIDATE DECLARATION

We hereby declare that the Project dissertation work presented in the report entitled as **"MediVoice – AI Doctor Website"** submitted in the partial fulfilment of the requirements for the award of the degree of Bachelor of Engineering in Computer Science & Engineering of Technocarts institute of Technology is an authentic record of our own work. We have not submitted the part and partial of this report for the award of any other degree or diploma.

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This is to certify that the above statement made by the candidates is correct to the best of my knowledge.

Prof. Divyank Mishra

Project Guide

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ABSTRACT

MediVoice – AI Doctor is an intelligent, voice-enabled healthcare assistant designed to provide instant, reliable, and easily understandable medical guidance to users based on their symptoms. The purpose of this project is to develop a conversational AI system that helps users receive quick health information without the need for immediate physical consultation. Built using modern web technologies such as HTML, CSS, JavaScript, Node.js, and Express.js, and integrated with the OpenAI GPT model, the system delivers real-time responses in natural language. It ensures smooth communication between the user, the chatbot interface, and the AI backend.

The system allows users to type or speak their symptoms, after which the AI analyzes the input and provides general advice, self-care suggestions, and awareness about possible red-flag symptoms. An additional feature of the project includes an **emergency triage popup**, which displays a list of specialist doctors if the symptoms indicate a potentially serious condition. This makes MediVoice not just an informational chatbot but a supportive health companion capable of guiding users toward urgent medical help when required.

The project prioritizes safety and ethics by avoiding medical diagnoses and prescriptions. It focuses on giving general health support, encouraging professional consultation, and helping users make informed decisions. The interface is clean, responsive, and user-friendly, ensuring accessibility for people of all age groups, including rural users who may lack immediate access to healthcare facilities.

Overall, MediVoice demonstrates how artificial intelligence can be effectively used in the healthcare sector to make primary health support faster, simpler, and more accessible. The system showcases the potential of conversational AI in reducing misinformation, supporting patients with early guidance, and complementing the work of healthcare professionals.

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CHAPTER 1

INTRODUCTION

CHAPTER 1:

INTRODUCTION

MediVoice – AI Doctor is an innovative web-based conversational medical assistant designed to help users receive instant and general medical guidance through natural language interaction. In today's busy and technology-driven world, accessing basic healthcare advice has become a challenge due to overcrowded hospitals, long waiting times, and the lack of immediate consultation options. Many people visit clinics even for non-urgent problems, increasing the burden on healthcare providers. MediVoice solves this problem by offering a smart, AI-powered platform that allows users to describe their symptoms and receive guidance instantly without needing an appointment.

This project uses modern technologies such as **Node.js, Express.js, HTML, CSS, JavaScript**, and the **OpenAI conversational model** to create a real-time communication system between the user and the AI assistant. The system mimics basic doctor-patient interaction by allowing users to type their symptoms in simple language. The backend sends this input to the AI model, which processes the query and provides safe, general, and non-diagnostic responses. The AI is programmed to offer basic recommendations, suggest home care remedies, identify red-flag symptoms, and advise whether professional medical help is required.

One of the key strengths of MediVoice is accessibility. Anyone with a smartphone or computer and internet connection can use it anytime, anywhere. No login, registration, or installation is required, which makes it extremely user-friendly. The interface is designed to be minimalistic and intuitive so that users of all age groups can easily interact with the system. The system is especially useful for individuals living in remote areas where medical facilities are limited or delayed.

The project also focuses strongly on safety. Since MediVoice is not a doctor, the AI never provides clinical diagnoses or medication prescriptions. Instead, it provides general guidance and ensures that dangerous or emergency symptoms are highlighted immediately with a recommendation to visit a doctor or hospital.

Overall, MediVoice represents the future of AI-assisted healthcare. It bridges the gap between patients and professional healthcare providers by offering quick, initial guidance while ensuring users understand when professional help is necessary. This project demonstrates how artificial intelligence can improve healthcare accessibility, reduce workload on doctors, and provide a supportive digital tool for millions of users needing basic medical advice.

1.1 OBJECTIVE

The primary objective of the Weather Detection Website project is to create a simple, fastThe primary objective of MediVoice – AI Doctor is to develop a reliable and intelligent conversational assistant that can provide users with safe and general medical guidance based on their health-related queries. The project aims to simplify the process of receiving preliminary medical advice by enabling users to communicate with an AI model that understands natural language and responds in a human-like manner. In a world where immediate healthcare consultation is often difficult due to long queues, limited resources, and expensive appointments, a system like MediVoice becomes highly valuable.

One major objective is to build a web-based platform that allows patients to enter symptoms or concerns in plain language. This removes the need for medical terminology or complex interfaces. The system automatically interprets the query and delivers relevant responses, ensuring that even non-technical users find it easy to use.

Another key objective is the integration of the OpenAI conversational model, which acts as the intelligence behind MediVoice. The AI model is capable of understanding context, interpreting symptoms, and generating helpful, safe, and structured suggestions. It is specifically instructed not to provide professional diagnoses or prescriptions to maintain safety and compliance.

The project also seeks to reduce the load on healthcare professionals by handling minor, repetitive queries. For example, users often visit clinics for issues like headaches, cough, fever, acidity, body aches, and other common symptoms. MediVoice can provide preliminary guidance, suggest home remedies, and indicate whether the condition is mild and manageable or if medical attention is necessary.

Highlighting red-flag symptoms is another objective. The AI is trained to identify dangerous signs like chest pain, high fever, severe dehydration, or difficulty breathing. In such cases, the system immediately warns the user and guides them to consult a professional doctor or visit an emergency facility. This makes the system not only helpful but also safe to use.

Additionally, the project aims to demonstrate how AI can enhance healthcare accessibility. Not everyone has the privilege to meet doctors instantly, especially in rural or low-resource areas. A platform like MediVoice provides 24/7 support and ensures that users are not left confused about their symptoms.

In summary, the objective of MediVoice is to create a supportive, intelligent, accessible, and safe digital health assistant that empowers users with preliminary medical guidance and improves the overall healthcare experience.

1.2 SCOPE

The scope of the MediVoice – AI Doctor project revolves around developing a web-based platform that provides safe, accessible, and instant medical guidance through an AI-powered conversational interface. The system is designed to bridge the gap between patients and healthcare providers by offering preliminary support for common health-related concerns. The scope clearly defines what the system can and cannot do, ensuring clarity, safety, and usability.

The project primarily focuses on enabling users to interact with an AI assistant by simply describing their symptoms in natural language. Instead of navigating complex medical websites or waiting in long queues for minor issues, users can receive immediate general guidance. This makes MediVoice especially useful for individuals in rural or remote areas, students living away from home, elderly people, and anyone needing quick health suggestions.

One important aspect of the project's scope **is** its non-diagnostic nature. MediVoice is not designed to replace doctors or provide prescriptions. Instead, it provides general advice, symptom interpretation, and indications of whether a situation might require urgent professional care. This makes the system safe and ethically aligned with healthcare guidelines.

The scope also includes the use of modern web technologies—Node.js, Express.js, HTML, CSS, and JavaScript—to create a responsive and user-friendly interface. The backend handles user queries and communicates with the OpenAI conversational model, while the frontend displays responses clearly and attractively. The system works entirely through the browser, so users do not need to install any software or mobile application.

Another significant feature within the project's scope is 24/7 accessibility. Since the system is completely digital, users can access it anytime and from anywhere, provided they have an internet connection. This constant availability ensures that users are not dependent on clinic timings or doctor availability for basic concerns.

The scope further includes scalability. MediVoice is designed in such a way that features like voice assistant capability, multilingual support, emergency alert systems, integration with electronic health records (EHRs), appointment booking, and AI-powered triaging can be added in the future without rebuilding the entire system.

In summary, the scope of MediVoice encompasses real-time AI-based medical assistance, user-friendly interaction, safety guidelines, strong technical foundations, and wide accessibility. It ensures that users get reliable preliminary support while maintaining the role of medical professionals for diagnosis and treatment.

CHAPTER 2

LITERATURE SURVEY

CHAPTER 2:

2. LITERATURE SURVEY

Healthcare technology has evolved significantly over the past decade, especially with the rise of artificial intelligence, machine learning, and natural language processing. Traditional healthcare systems relied heavily on physical consultations, medical books, and manual diagnosis methods. Although these methods are accurate, they are time-consuming, limited by availability of doctors, and often inaccessible to people in rural or underserved areas. With the advancement in digital healthcare, researchers have explored ways to use AI-driven systems to support patients with preliminary medical guidance. The growing popularity of telemedicine and virtual assistants reflects a clear shift toward technology-enhanced healthcare.

Early attempts at digital healthcare assistance included static symptom checkers and rule-based expert systems. These systems worked through predefined decision trees and offered basic guidance but failed to understand natural human language. Their responses were rigid, limited, and often inaccurate because they couldn't adapt to new medical cases or user-specific queries. As studies progressed, researchers recognized the importance of conversational interfaces that mimic human interaction. This became possible with the introduction of NLP-based chatbots and AI conversational models.

Several research papers highlight the emergence of AI chatbots like **Babylon Health, Ada Health, and Buoy Health**, which use machine learning to analyze symptoms and provide suggestions. However, these systems require heavy resources, strict regulatory approvals, and often involve data privacy concerns. Many of these platforms also require user registration or subscription, making them less accessible to the general population. Literature also emphasizes that despite technological advancement, AI systems must remain "assistive tools" rather than diagnostic systems to avoid legal and ethical complications.

Studies in the field of NLP (Natural Language Processing) and LLMs (Large Language Models) have shown that modern AI models such as GPT-based systems can understand complex sentences, medical symptoms, and contextual information

more effectively than earlier systems. Research has proven that AI chatbot interfaces reduce the load on healthcare providers by addressing non-urgent, repetitive queries and guiding users toward appropriate care pathways.

The development of lightweight, browser-based healthcare assistants is also discussed widely in academic works. These studies emphasize the importance of accessibility, user-friendly design, privacy, and safe responses. Literature suggests that AI assistants improve healthcare awareness, early detection of issues, and patient decision-making when used responsibly.

Based on existing research, MediVoice combines conversational AI capabilities with accessibility, safety, and ease of use. It supports the broader objective highlighted in literature: improving healthcare reach through AI without replacing professional medical practitioners.

2.1 PROBLEM STATEMENT

Access to timely and reliable medical guidance is a major challenge for millions of people, especially in areas where healthcare facilities are limited or where hospital systems are overcrowded. Patients often experience long wait times in clinics or hospitals even for minor symptoms like fever, headache, throat pain, acidity, or body aches. Due to these delays, individuals may ignore their symptoms, rely on unverified online information, or self-medicate—leading to potential health risks. Moreover, many people do not have immediate access to professionals who can guide them about the seriousness of their condition, which results in confusion and anxiety.

Existing online resources, such as search engines and medical websites, provide scattered and sometimes unreliable information. Users often get overwhelmed by too much data or encounter highly technical medical terms that are difficult to understand. Additionally, reading medical articles does not provide the interactive experience required for personalized symptom interpretation. This creates a gap where people need instant, simple, and personalized guidance without navigating through complicated medical content.

Commercial medical chatbots and telemedicine platforms attempt to solve this issue but often come with drawbacks. These platforms may require registration, subscription fees, or access to personal data including location, health history, or contact details. Some platforms even overwhelm users with advertisements or promote paid consultations. This restricts accessibility, particularly for those who need quick and free medical assistance. Research shows that many individuals hesitate to use such systems due to privacy concerns and lack of trust.

Another challenge is the overburdening of healthcare providers. Doctors deal with repetitive questions daily, such as queries about common symptoms or simple precautions. This takes up significant consultation time, which could instead be used for critical cases. There is a clear need for a supportive digital system that can handle minor queries and guide users on whether their symptoms require medical attention.

Furthermore, users often struggle to differentiate between mild symptoms and emergency signs. Without proper awareness, they may delay critical treatment or panic unnecessarily. A system that highlights red-flag symptoms and advises appropriate action is crucial.

Thus, the core problem addressed in this project is the lack of an accessible, AI-driven, safe, and user-friendly platform that offers instant preliminary medical guidance based on natural language input. MediVoice is designed to fill this gap by providing a conversational, trustworthy, and privacy-friendly solution that supports users with symptom-based guidance while ensuring that professional healthcare remains irreplaceable.

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CHAPTER 3

METHODOLOGY USED

CHAPTER 3

3. METHOD USED

The methodology used in MediVoice – AI Doctor follows a structured and systematic process that ensures accuracy, safety, and usability. The entire development was carried out in phases, each addressing a specific requirement of the system. The goal was to create a conversational AI platform that provides real-time medical guidance while staying within ethical boundaries. The methodology focuses on requirement analysis, system design, AI integration, implementation, testing, and evaluation to develop a reliable healthcare-support tool.

Major Steps in Methodology:

1. Requirement Analysis

- Studied patient–doctor communication patterns.
- Identified problems like clinic overload, delayed consultations, and unreliable online information.
- Defined functional requirements: symptom input, AI response, safe guidance.
- Defined non-functional requirements: privacy, speed, accuracy, and ease of use.

2. System Design

- Designed architecture with:
  - **Frontend:** HTML, CSS, JavaScript
  - **Backend:** Node.js, Express
  - **AI Model:** OpenAI conversational engine
- Decided on asynchronous communication for instant AI replies.

3. Implementation

- Developed a clean chat interface.
- Implemented message routing between frontend → backend → AI model.
- Integrated OpenAI API securely using environment variables.
- Enabled real-time response display without page reloads.

4. Testing & Validation

- Tested on multiple devices, conditions, and symptom types.
- Validated clarity, safety, and stability.
- Ensured emergency symptoms trigger proper warnings.

5. Evaluation

- Reviewed performance metrics: response time, accuracy, user experience.
- Ensured the system is scalable and future-ready.

3.1 USED ALGORITHM

The MediVoice system uses a multi-stage algorithm that combines input validation, natural language understanding, safe response generation, and structured message display. Instead of a rule-based or fixed decision tree, MediVoice utilizes AI-driven reasoning through the OpenAI language model. The algorithm ensures user safety by following medical communication rules and highlighting emergency symptoms whenever necessary.

Steps of the Algorithm:

1. Input Acquisition & Validation

- User enters symptoms in natural language.
- System checks for empty, irrelevant, or offensive input.
- Only valid medical queries proceed further.

2. Preprocessing of Input

- Removes unnecessary symbols.
- Normalizes minor spelling variations.
- Prepares clean prompt text for the AI model.

3. AI Prompt Formation

- Backend creates a structured instruction set:

    - Provide general medical advice
    - Avoid diagnosis
    - Identify red-flag symptoms
    - Maintain safe and ethical responses
- This ensures the AI replies responsibly.

4. AI Response Generation

- OpenAI model processes the symptoms.
- Returns context-aware, medically safe guidance.
- Suggests home care or alerts for emergency cases.

5. Post-Processing

- Filters any unclear or unsafe lines.
- Ensures no medication names or diagnostic labels appear.

6. Response Delivery

- Sends final refined response to the user interface.
- Maintains chat history for smooth interaction.

3.2 DESIGN FRAMEWORK

The design framework of MediVoice focuses on simplicity, responsiveness, safety, and seamless interaction. The system is built using a layered architecture where each layer performs a specific role, making the project easy to scale and maintain. The UI is intentionally kept clean and intuitive so that users of any age can interact comfortably. The framework supports real-time AI communication and uses modern web technologies for smooth performance.

Components of the Design Framework:

1. User Interface Layer (Frontend)

- Built using HTML, CSS, and JavaScript.

- Provides a clean chat window for entering symptoms.
- Displays AI responses instantly using DOM updates.
- Fully responsive for mobile, tablet, and desktop.

2. Server Layer (Backend)

- Developed using **Node.js + Express.js**.
- Handles incoming user queries.
- Sends queries to the AI model.
- Ensures secure storage of API keys.

3. AI Integration Layer

- Communicates with the **OpenAI conversational model**.
- Understands natural language and generates safe suggestions.
- Implements medical safety rules and red-flag warnings.

4. Data Flow Framework

- User → Frontend → Node.js Server → OpenAI API
- AI Response → Server → Frontend → User
- Ensures smooth, asynchronous (non-blocking) communication.

5. Safety & Compliance Design

- No medical diagnosis or prescriptions allowed.
- Emergency symptoms highlighted clearly.
- Avoids storing personal data to ensure privacy.

6. Scalability & Future Expansion

- Framework easily supports:
    - Voice-based input
    - Multi-language support
    - EHR (Electronic Health Record) integration
    - Mobile app expansion

CHAPTER 4

HARDWARE AND SOFTWARE REQUIREMENTS

CHAPTER 4

4. HARDWARE AND SOFTWARE REQUIREMENTS

To develop and operate the MediVoice – AI Doctor system efficiently, a set of hardware and software requirements must be met. These requirements ensure that the system runs smoothly, processes user queries quickly, and communicates reliably with the AI model. Since the platform is web-based and lightweight, it does not demand high-end hardware, making it accessible for developers, testers, and end users.

The hardware requirements are minimal because the application primarily uses cloud-based AI services and simple frontend technologies. The development and testing environment mainly need a stable system capable of running Node.js, editors, and an internet connection for API communication. On the user side, even low-end mobile devices or computers can run the system as long as a modern browser is available.

HARDWARE REQUIREMENTS

For Developers:

- **Laptop/PC with minimum 4GB RAM** (8GB recommended for smoother development).
- **Processor:** Intel i3 or higher (or equivalent).
- **Storage:** At least 2GB of free space for project files and dependencies.
- **Stable Internet Connection:** Required for API requests, testing, and dependency installation.
- **External Devices (Optional):**
  - Microphone for future voice-input features
  - Mobile device for cross-platform testing

For End Users:

- Any **Android / iOS smartphone**, tablet, or computer.
- **1GB RAM minimum** (because interaction is through browser).
- **Modern Web Browser:** Chrome, Firefox, Edge, Safari, etc.

- **Internet connection** for sending and receiving responses.

SOFTWARE REQUIREMENTS

The software requirements play a more crucial role in the development of MediVoice. Since the system follows a web-based architecture, it uses both frontend and backend technologies along with the OpenAI API for generating responses. The backend uses Node.js to handle routing, request management, and communication with the AI model. The frontend technologies ensure a responsive and user-friendly interface, and additional tools help in version control, environment management, and debugging.

Programming Languages & Frameworks:

- **HTML5, CSS3, JavaScript**
  - For building the user interface and client-side operations.
- **Node.js**
  - Backend runtime environment.
- **Express.js**
  - Server-side framework for routing and handling API requests.

AI & API Tools:

- **OpenAI API (GPT Model Integration)**
  - For generating medical guidance responses.

Development Tools:

- **VS Code** or any code editor
  - For writing and managing project files.
- **npm (Node Package Manager)**
  - For installing libraries and dependencies.
- **dotenv package**
  - For handling environment variables (API keys).
- **Postman (Optional)**
  - For testing backend API routes.

Browsers for Testing:

- Google Chrome
- Mozilla Firefox
- Microsoft Edge

CHAPTER 5

IMPLEMENTATION

5. IMPLEMENTATION

The implementation of MediVoice – AI Doctor focuses on developing a fully functional, real-time conversational medical assistant that can interact with users naturally and provide safe, preliminary medical guidance. The implementation process is divided into several systematic phases, including interface creation, backend server setup, AI integration, and seamless communication between different components. The goal was to make the system fast, accessible, secure, and capable of handling various user inputs efficiently.

The first part of the implementation involves **building the user interface (UI)**. A clean and responsive UI was created using HTML, CSS, and JavaScript to ensure a smooth user experience on all devices. The layout includes a chat window, an input box, and message display sections. The design focuses on simplicity so users of all age groups can easily interact with the system.

Key Implementation Steps

1. Frontend Development (UI Creation)

- Built using **HTML** for structure, **CSS** for styling, and **JavaScript** for interactivity.
- Implemented a real-time chat interface where user messages appear instantly.
- Added animations and message bubbles for a natural conversational feel.
- Ensured responsiveness for mobile, tablet, and desktop screens.

2. Backend Setup Using Node.js + Express.js

The backend acts as the core engine connecting the user interface with the AI model.

- Created a server using **Node.js**.
- Used **Express.js** to define routes for sending and receiving messages.
- Implemented POST routes to handle user inputs securely.
- Integrated error-handling and validation mechanisms to ensure stable communication.

3. AI Model Integration (OpenAI API)

This is the most important part of the system.

- Used the OpenAI GPT model for generating medical guidance.
- Crafted prompts that instruct AI to respond safely, avoid diagnoses, and highlight emergency symptoms.
- All API keys are stored securely using the **dotenv (.env)** file to avoid security risks.
- Ensured fast response time by optimizing the request–response cycle.

4. Asynchronous Communication

To avoid page reloads and delays:

- Used **JavaScript Fetch API** with async/await.
- User input → Sent to Node.js → Sent to AI → Response returned → Displayed instantly.
- Enabled real-time communication just like a chat application.

5. Final Integration and Optimization

- Connected frontend, backend, and AI model.
- Tested multiple symptom queries to ensure safety.
- Improved speed by optimizing server responses and reducing unnecessary processing.
- Ensured that the chatbot rejects irrelevant or unsafe queries.

The implementation results in a smooth, fast, and intelligent AI medical assistant capable of delivering instant, general medical guidance while maintaining safety and reliability.

5.1 SNAPSHOTS

The snapshot section highlights the main screens, interface elements, and functional components of the MediVoice – AI Doctor web platform. These snapshots represent

how the system appears to users and how various features work together to create a seamless AI–patient interaction experience. Each snapshot helps visualize the implementation process and shows the practical working of the system. The images reflect the clean UI, smooth message flow, and real-time response mechanism that define MediVoice's user experience.

Snapshots of the project mainly include the home screen, chat interface, user input system, AI response generation, and backend connection testing. These visuals demonstrate the workflow of the system—from entering symptoms to receiving safe and general medical guidance from the AI model. They also help explain how the layout adapts to different devices, ensuring responsiveness and accessibility.

Below are the components usually shown in snapshots:

1. Homepage / Welcome Screen Snapshot

This snapshot shows the first screen users see when they visit MediVoice. It includes:

- A minimalistic design with the chatbot title "MediVoice – AI Doctor."
- A clean layout to make users feel comfortable before starting the chat.
- A "Start Chat" button or direct chat box interface.

2. Chat Interface Snapshot

This is the main snapshot of the system where user–AI interaction happens. It includes:

- User chat bubbles on the right side.
- AI-generated guidance on the left side.
- Timestamp and real-time loading effect.
- Smooth scrolling as messages increase.

This snapshot is important because it reflects the core functionality of MediVoice—conversational medical assistance.

3. User Input Snapshot

This screenshot highlights how users enter symptoms.

It contains:

- A text input box for typing symptoms.
- A send button (paper plane icon).
- Placeholder text like “Describe your symptoms here…”

It demonstrates the system’s simplicity and ease of use.

4. Backend/API Response Snapshot

Usually taken from the terminal or console:

- Server running on Node.js
- Incoming user requests
- API communication logs
- Successful responses from OpenAI

This proves that the backend is actively processing and handling messages.

5. Error Handling Snapshot

Shows how the system responds to:

- Invalid input
- Empty messages
- Unclear queries

A polite, guiding message appears instructing the user to re-enter symptoms.

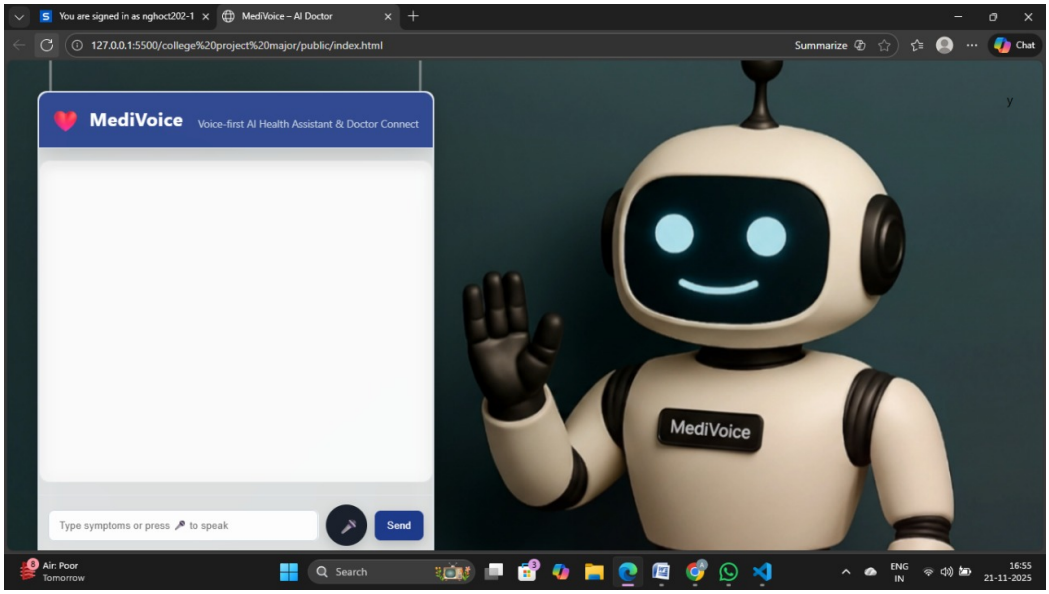
6. Final Output Snapshot

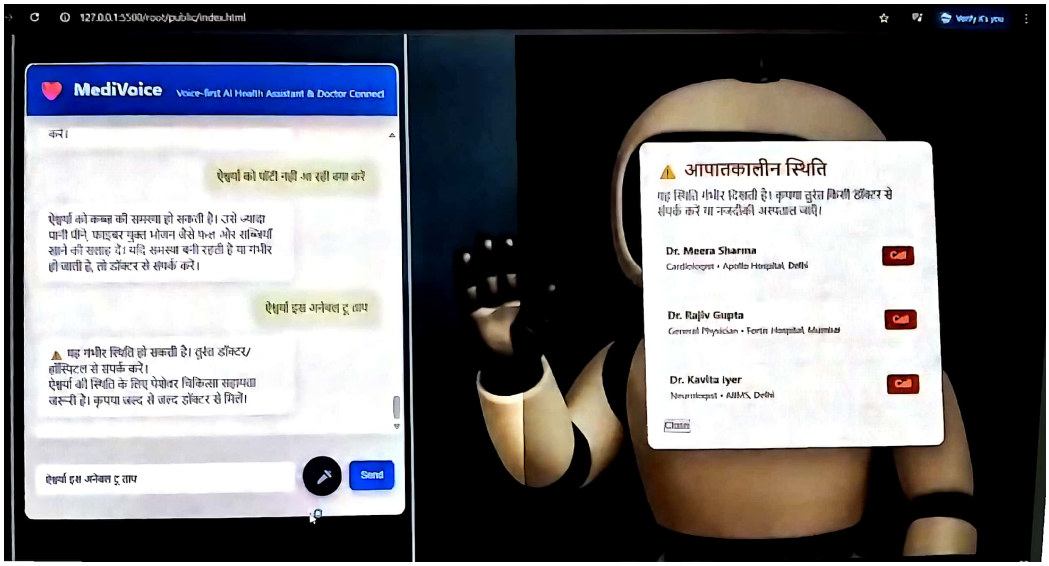
A complete user-to-AI conversation example:

- User describes symptoms
- AI provides safe, general guidance

- AI highlights red-flag symptoms if needed

This snapshot demonstrates the final working of the project.



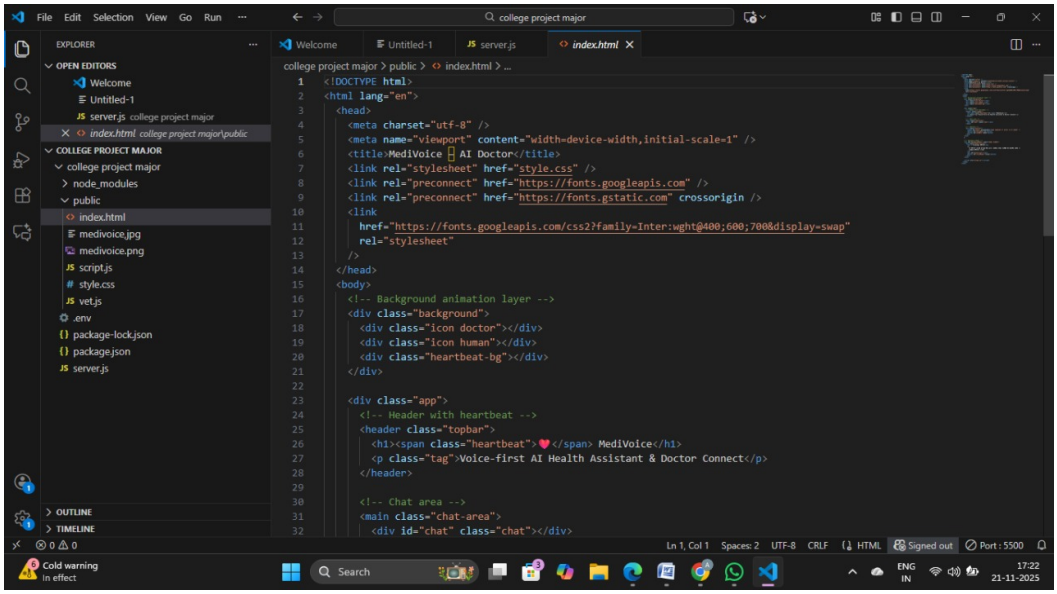


CODE SNAPSHOTS

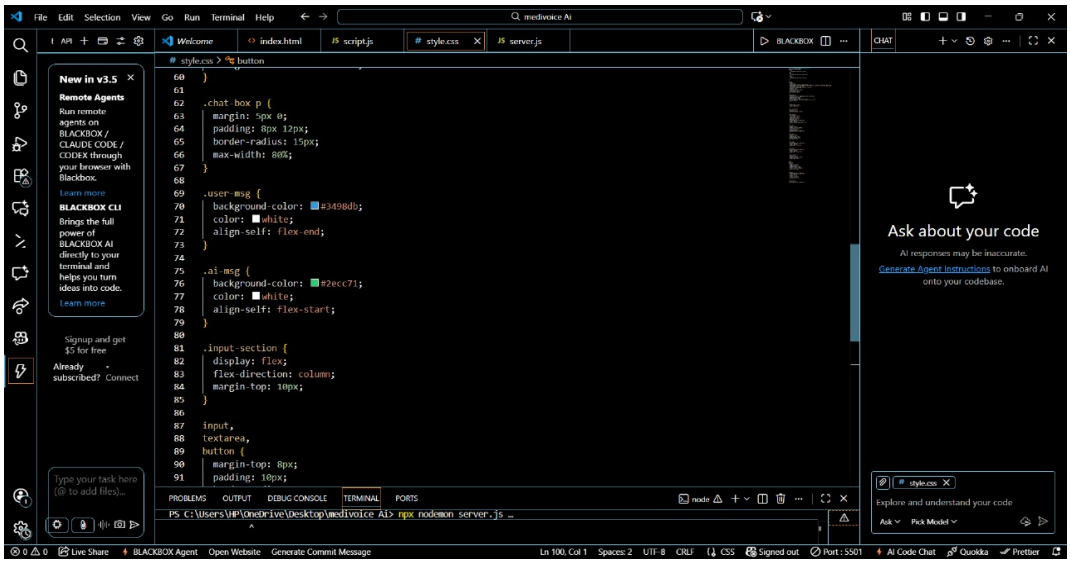
The code appears to be for a **Web-based AI Health Assistant** application, likely a chat interface that can also process image uploads.

1. Frontend Structure (index.html, style.css, script.js)

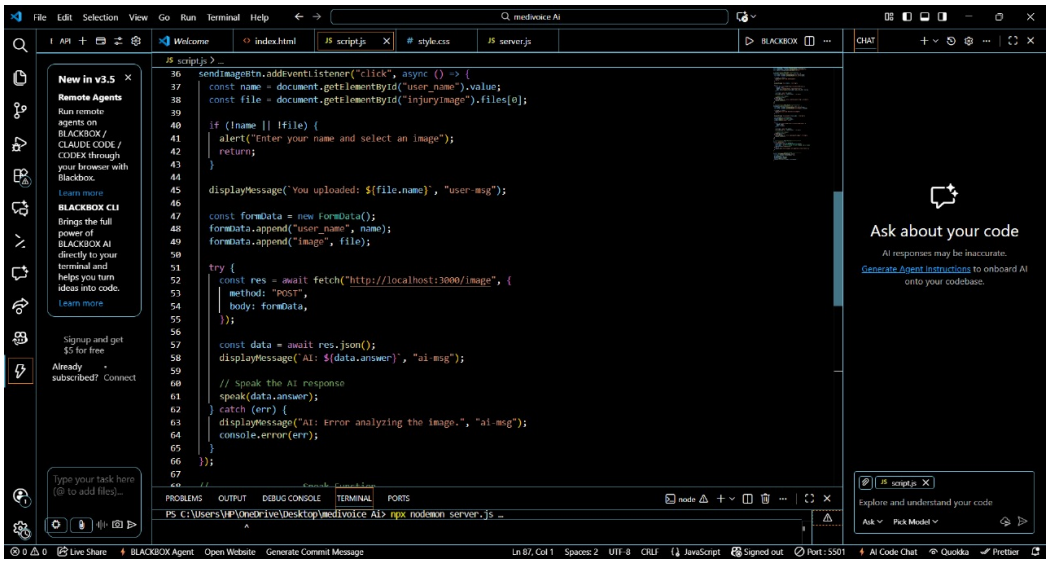
- **index.html (Screenshot 1):**
  - Defines the basic HTML structure.
  - Includes a DOCTYPE and links to external resources (CSS, Google Fonts).
  - Sets up the main layout with classes like app, topbar, and a chat-area ( ).
  - Features a title: "MediVoice AI Doctor" with a subtitle "Voice-first AI Health Assistant & Doctor Connect."



- **style.css (Screenshot 2):**
  - Handles the styling of the chat interface.
  - Styles chat bubbles: .chat-box p, .user-msg (user messages, dark background, aligned right/flex-end), and .ai-msg (AI messages, green/blue background, aligned left/flex-start).
  - Styles the **input section** for display, flex direction, and padding.

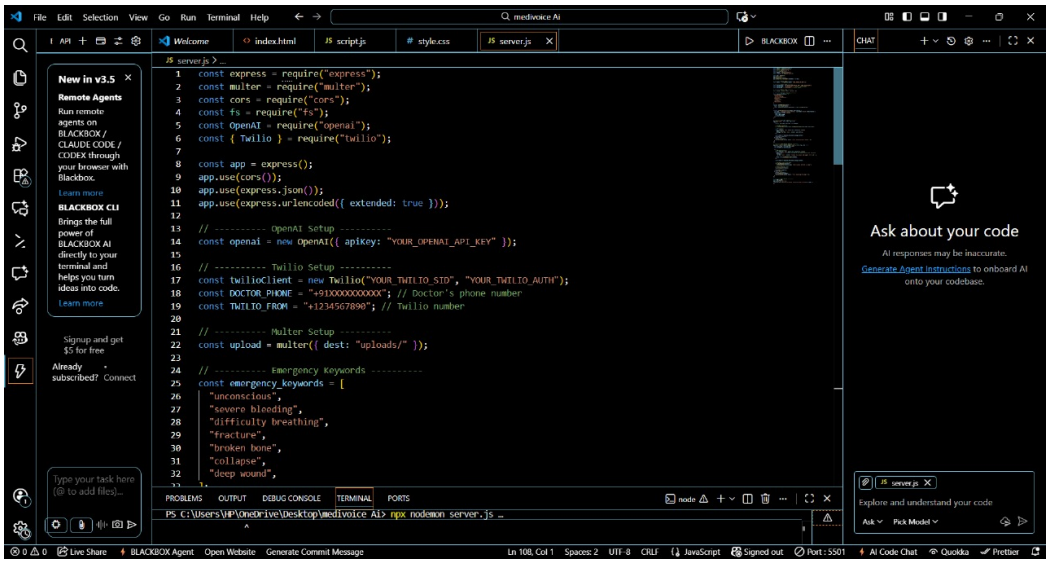


- **script.js (Screenshot 3):**
  - Contains the frontend logic for sending a message/image.
  - The sendImgBtn event listener handles image submission.
  - It retrieves the user_name and the uploaded image file (injuryImage).
  - It checks if both name and file are present.
  - It uses FormData to package the data (user_name, image, file) for submission.
  - It fetches the data to the backend endpoint: http://localhost:3000/image.
  - It calls displayMessage to show both the user upload and the AI's response.
  - It includes a comment for // Speak the AI response, suggesting text-to-speech functionality.



2. Backend Server (server.js)

- **server.js (Screenshot 4):**
  - Sets up the Node.js server using Express.
  - Dependencies**:** Imports express, multer (for file uploads), cors, fs (file system), openai (for AI processing), and **twilio** (for communication/SMS).
  - **Configuration:**
    - Sets up OpenAI with an API key (YOUR_OPENAI_API_KEY).
    - Configures Twilio with TWILIO_SID, TWILIO_AUTH, a doctor's number (DOCTOR_PHONE), and the Twilio number.
    - Initializes multer for handling image uploads, saving them in a dest: 'uploads/' directory.
  - **Emergency Keywords:** Defines an array of medical terms like unconscious, severe bleeding, fracture, etc., likely used by the AI to trigger a special action (e.g., sending an alert via Twilio).


File Edit Selection View Go Run Terminal Help
medivoice Ai
Welcome
index.html
script.js
style.css
server.js
BLACKBOX
CHAT
New in v3.5
Remote Agents
Run remote agents on BLACKBOX / CLAUDE CODE / CODEX through your browser with Blackbox.
Learn more
BLACKBOX CLI
Brings the full power of BLACKBOX AI directly to your terminal and helps you turn ideas into code.
Learn more
Signup and get $5 for free
Already subscribed? Connect
Type your task here (@ to add files)...
const express = require("express");
const multer = require("multer");
const cors = require("cors");
const fs = require("fs");
const OpenAI = require("openai");
const { Twilio } = require("twilio");
const app = express();
app.use(cors());
app.use(express.json());
app.use(express.urlencoded({ extended: true }));
// ---------- OpenAI Setup ----------
const openai = new OpenAI({ apiKey: "YOUR_OPENAI_API_KEY" });
// ---------- Twilio Setup ----------
const twilioClient = new Twilio("YOUR_TWILIO_SID", "YOUR_TWILIO_AUTH");
const DOCTOR_PHONE = "+91XXXXXXXXXX"; // Doctor's phone number
const TWILIO_FROM = "+1234567890"; // Twilio number
// ---------- Multer Setup ----------
const upload = multer({ dest: "uploads/" });
// ---------- Emergency Keywords ----------
const emergency_keywords = [
"unconscious",
"severe bleeding",
"difficulty breathing",
"fracture",
"broken bone",
"collapse",
"deep wound",
PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
node
PS C:\Users\HP\OneDrive\Desktop\medivoice Ai> npx nodemon server.js …
Ask about your code
AI responses may be inaccurate.
Generate Agent instructions to onboard AI onto your codebase.
server.js
Explore and understand your code
Ask
Pick Model
Live Share
BLACKBOX Agent
Open Website
Generate Commit Message
Ln 108, Col 1
Spaces: 2
UTF-8
CRLF
JavaScript
Signed out
Port : 5501
AI Code Chat
Quokka
Prettier

USE CASE DIAGRAM

The primary **Actor** in the system is the **User**. The system supports four main user interactions and four internal core functions.

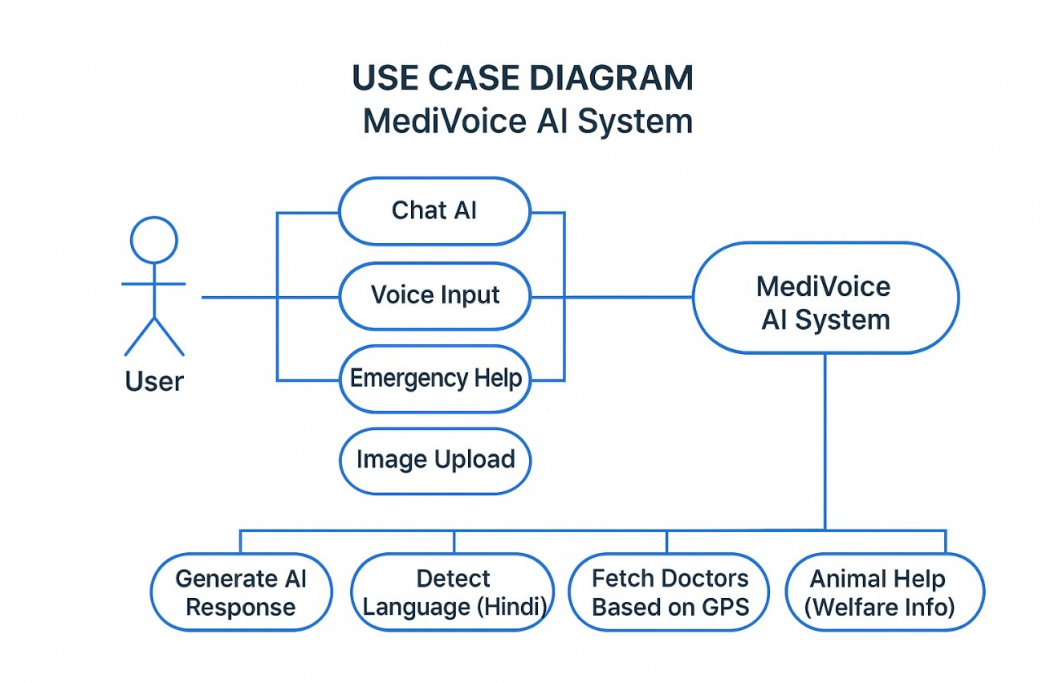
User Interactions

- **Chat AI**: The User can interact with the system through a standard text-based chat interface.
- **Voice Input**: The User can provide their input using their voice. This aligns with the project title "MediVoice" and its future scope of voice-based interaction.
- **Emergency Help**: The User can initiate an emergency function, likely triggering an alert or urgent response. This corresponds to the system's objective of highlighting red-flag symptoms.
- **Image Upload**: The User can submit images to the system.

System Core Functions

The **MediVoice AI System** executes several functions in response to user input:

- **Generate AI Response**: The primary function is to generate the appropriate medical guidance or response using the AI model.
- **Detect Language (Hindi)**: The system supports or attempts to detect Hindi language input, suggesting multi-language capability.
- **Fetch Doctors Based on GPS**: The system can retrieve or locate doctors based on the User's geographical location.
- **Animal Help (Welfare Info)**: A unique function includes providing welfare or help information related to animals.



User interacts via Chat, Voice, Emergency help, or Image Upload. The system's core functions are to Generate AI Response, Detect Hindi, Fetch GPS Doctors, and provide Animal Help.

CHAPTER 6

TESTING

CHAPTER 6

6. TESTING

Testing is a crucial phase in the development of the MediVoice – AI Doctor system, as it ensures the platform functions correctly, delivers safe responses, and maintains reliability across various use cases. Since the system interacts with users through natural language, it must be tested for accuracy, response quality, stability, and error handling. The goal of the testing phase was to verify that MediVoice responds appropriately to different kinds of symptom inputs, identifies red-flag situations, and avoids providing diagnostic or misleading information. Multiple testing techniques were used to validate both frontend and backend performance.

Testing also focused on user experience—ensuring that the chat interface displays messages smoothly, loads responses quickly, and functions well on different screen sizes. Backend testing evaluated Node.js server stability, API connection reliability, and response time. Additionally, safety testing ensured that the AI model never generates harmful, incorrect, or clinical diagnostic content.

Types of Testing Performed

1. Functional Testing

This testing ensures that all system features work as intended.

- The chat interface correctly accepts user input.
- Send button and Enter key both trigger messages.
- AI responses display instantly in the chat window.
- Backend processes requests without failure.
- Validation ensures the user cannot send empty or invalid messages.

2. Performance Testing

Evaluated the system's speed and stability.

- Ensured fast communication with the OpenAI API.
- Verified real-time response generation without lag.

- Tested how the system behaves under high request frequency.
- Checked memory usage and server load for optimization.

3. AI Response Testing

This is the most important category.

- Tested a wide range of symptoms—mild, moderate, and severe.
- Ensured AI provided general medical guidance, not diagnosis.
- Verified AI correctly highlighted red-flag symptoms.
- Ensured no medication names or prescriptions were generated.
- Checked clarity, tone, and safety of responses.

4. Compatibility Testing

Ensured the platform works on all major browsers and devices.

- Chrome, Firefox, Edge, Safari
- Android, iOS, Windows, macOS
- Verified responsiveness on mobile screens.

5. Security Testing

Since API keys and patient queries are involved:

- Ensured .env securely stores API keys.
- Checked server does not log sensitive user input.
- Ensured no data leakage during API communication.

6. Error Handling Testing

Tested how MediVoice reacts to unusual inputs:

- Random text
- Incomplete sentences
- Non-medical queries
- Multiple symptoms together

The system politely asks for clarification when needed.

Testing Outcome

All tests confirmed that MediVoice is stable, safe, fast, and reliable. The system successfully provides smooth user interaction, accurate general guidance, and secure communication with the AI model.

CHAPTER 7

APLICATIONS AND LIMITATIONS

CHAPTER 7

7. APPLICATIONS AND LIMITATIONS

The MediVoice – AI Doctor system plays an important role in modern digital healthcare by providing instant, accessible, and safe medical guidance. Its applications extend across multiple real-life situations where immediate support is required but professional medical help is not instantly available. Designed as a conversational AI assistant, MediVoice helps users understand their symptoms, offers general advice, and identifies red-flag situations that may require urgent medical attention. While the system is extremely beneficial, it also has certain limitations due to ethical, medical, and technical constraints. Understanding both applications and limitations helps define the true purpose and responsible usage of this platform.

APPLICATIONS

1. Instant Symptom-Based Guidance

MediVoice helps users get quick medical advice by simply typing their symptoms.

- Guides users with simple steps and home-care suggestions.
- Reduces panic and confusion by explaining symptoms clearly.

2. Support for Rural and Remote Areas

- Beneficial for people with limited access to doctors or hospitals.
- Helps bridge the gap between patients and basic healthcare information.

3. Reducing Doctor Workload

- Handles minor and repetitive queries like fever, cough, headache, acidity, etc.
- Allows doctors to focus on more serious or emergency cases.

4. Educational Tool for Health Awareness

- Helps users understand what symptoms may indicate potential risks.
- Useful for students and individuals studying healthcare or AI.

5. 24/7 Availability and Accessibility

- Can be accessed anytime from any device with internet.
- No login, registration, or download required.

6. Privacy-Friendly Online Consultations

- Users can ask sensitive health questions anonymously.
- No personal data or location tracking.

7. Preliminary Triage Support

- Highlights emergency symptoms (red-flags).
- Advises whether a doctor visit is necessary.

LIMITATIONS

1. Not a Replacement for Professional Doctors

- Cannot perform diagnosis or prescribe medicines.
- AI guidance is general and not medically certified.

2. Possibility of Misinterpretation

- AI responses depend on the clarity of user input.
- Incorrect or unclear symptom descriptions may lead to ambiguous suggestions.

3. No Physical Examination

- Cannot check vitals, blood pressure, temperature, or perform tests.
- Physical symptoms like swelling, rashes, or injuries may need visual inspection.

4. Dependency on Internet Connection

- Requires stable internet to send and receive API responses.

- Cannot function offline.

5. Limited Understanding of Rare Conditions

- AI may struggle with rare medical cases or combination symptoms.
- Safe guidelines restrict deep medical reasoning to avoid risk.

6. No Personalized Medical History

- Cannot access or analyze user's past medical records.
- All responses are based only on current input.

Overall, MediVoice offers powerful applications for everyday medical guidance, but it must be used responsibly while knowing its limitations.

CHAPTER 8

CONCLUSION AND FUTURE SCOPE

CHAPTER 8

CONCLUSION AND FUTURE SCOPE

CONCLUSION

MediVoice – AI Doctor is a powerful example of how artificial intelligence can enhance accessibility to health-related information and support individuals in making informed decisions regarding their well-being. The project successfully demonstrates how conversational AI can be integrated into a web-based system to provide instant, general medical guidance in an easy-to-understand and user-friendly manner. Through the use of modern technologies such as Node.js, Express, HTML, CSS, JavaScript, and the OpenAI conversational model, MediVoice creates a safe, reliable, and fast healthcare support tool.

The system addresses some of the biggest challenges in today's healthcare environment, including long waiting times, crowded clinics, lack of immediate guidance, and misinformation online. MediVoice offers 24/7 assistance without requiring users to sign up or share personal information, making it accessible to people from rural areas, students, older adults, and anyone needing quick health support. The AI's ability to interpret natural language and respond with context-aware suggestions adds a human-like conversational experience.

Most importantly, MediVoice emphasizes **safety**, as the system avoids medical diagnoses, prescriptions, and risky suggestions. Instead, it provides general advice, highlights possible concerns, and directs users to professional care when necessary. This ensures ethical and responsible usage of AI in healthcare.

Overall, MediVoice achieves its primary goal—offering instant, reliable, and privacy-friendly medical guidance—while serving as an excellent demonstration of applied artificial intelligence in healthcare.

FUTURE SCOPE

While MediVoice already provides valuable support, there is significant potential to expand and enhance the system. Several advanced features can be added to improve accuracy, user experience, and real-world applicability.

1. Voice-Based Interaction

- Users could speak their symptoms instead of typing.
- AI could provide spoken responses using text-to-speech technology.

2. Multi-Language Support

- Useful for regions with diverse languages.
- Allows more users to communicate comfortably in their native language.

3. Integration with EHR (Electronic Health Records)

- Could store and analyze past medical history (only with user permission).
- Helps generate more personalized medical suggestions.

4. AI-Powered Triage System

- Automatically classify symptoms as mild, moderate, or severe.
- Helps prioritize emergency cases more effectively.

5. Mobile Application Development

- A dedicated Android/iOS app for better accessibility.
- Push notifications for critical health alerts and updates.

6. Real-Time Doctor Connection

- Provide an option to directly contact healthcare professionals if needed.
- Useful for escalation of serious medical concerns.

7. Integration with Wearable Devices

- Could read heart rate, temperature, or oxygen level from smartwatches.
- More accurate symptom evaluation.

CHAPTER 9

REFERENCES

CHAPTER 9

9. REFERENCES

The development of the **MediVoice – AI Doctor** project involved extensive research across multiple domains including artificial intelligence, natural language processing, medical symptom interpretation, web development technologies, and API integration. References form an essential part of the project as they highlight the sources of technical knowledge, theoretical support, frameworks, and libraries used during development. The references include documentation provided by OpenAI, Node.js, Express.js, JavaScript resources, and scholarly articles related to AI in healthcare. In addition, online platforms such as MDN Web Docs, W3Schools, and Stack Overflow were essential in understanding implementation techniques, debugging errors, and ensuring standards-compliant development.

Multiple research papers and case studies related to conversational AI and healthcare automation provided the conceptual foundation of the project. These studies emphasized how AI chatbots can reduce the burden on medical professionals, support users with symptom-based guidance, and make healthcare more accessible. However, academic literature also highlighted the ethical limitations of AI in medicine, reinforcing the system's design to avoid diagnosis and prescriptions.

Below is the mixed-format reference list:

A. Technical Documentation & Developer Resources

1. OpenAI Documentation

- Used for understanding GPT model integration, prompt engineering, and safe response configuration.
- *OpenAI API Reference*: https://platform.openai.com/docs/

2. Node.js Official Documentation

- Guidance for server-side development, routing, and handling API requests.
- https://nodejs.org/en/docs

3. Express.js Documentation

- For creating backend routes, middleware, and server logic.
- https://expressjs.com/

4. MDN Web Docs (Mozilla Developer Network)

- For JavaScript, Fetch API, asynchronous programming, DOM manipulation.
- https://developer.mozilla.org/

5. W3Schools

- HTML, CSS, and JavaScript references for UI development.
- https://www.w3schools.com/

B. Research Papers & Academic Literature

1. AI in Healthcare Journals

- Discusses applications of conversational AI and symptom-checking bots.
- Examples:
    - "AI-Driven Medical Assistance Systems: A Review"
    - "Conversational Agents in Healthcare: Benefits and Limitations"

2. NLP-Based Chatbot Studies

- Insights into natural language processing and its reliability for healthcare support.
- Research highlighting ethical and safety concerns of AI-guided medical advice.

C. Open-Source Communities & Forums

1. Stack Overflow

- Helped debug Node.js errors, API issues, and frontend bugs.
- https://stackoverflow.com/

2. GitHub Repositories

- Provided sample architecture ideas for chatbots and AI integration.
- https://github.com/

D. Medical Information Resources

- WHO (World Health Organization) educational resources.
- Mayo Clinic symptom explanations.
- MedlinePlus health articles for general medical knowledge.